

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 2 – SCENARIO BASED ASSESSMENT

Course Code:	CSA06	Course Title:	Design and Analysis of Algorithms
CO Assessed:	CO2 – Manipulation (BL1, BL2, BL3)	Assessment:	Assessment Tool 2 – Scenario Based Assessment
Weightage:	20% of CIA	Total Marks:	100
CO2 Statement:	<i>Apply Brute Force technique to solve sorting, searching, Closest-Pair and Convex-Hull problems (BL1, BL2, BL3)</i>		
Student Name:	Y Greeshma Reddy	Reg. No.:	192424146
Section / Batch:	B Tech-AI&DS	Date:	23-07-2026

Instructions to Students

- Answer BOTH scenario-based questions. Each question carries 50 Marks. Total: 100 Marks.
- Carefully analyze the given scenario and identify the computational problem involved.
- Apply appropriate Brute Force techniques for sorting, searching, Closest-Pair, and Convex-Hull problems wherever applicable.
- Clearly justify the chosen solution approach with proper reasoning and analytical interpretation.
- Demonstrate decision-making skills by comparing possible solutions and selecting the most suitable approach.
- Use diagrams, stepwise illustrations, or algorithmic representations wherever necessary.
- Maintain clarity, logical organization, and proper technical presentation throughout the answers.

Question Type	Focus Area	Questions	Marks
Scenario Based Assessment	Analyze real-world scenarios and apply Brute Force techniques for sorting, searching, Closest-Pair, and Convex-Hull problems	Q1 - Q2	100 (2 × 50)
GRAND TOTAL:		100 Marks	

SECTION A – SCENARIO BASED ASSESSMENT

Q1. A library system stores book IDs in an unsorted list due to frequent additions and deletions. Users search for books using a basic search feature.

- Explain how Linear Search is applied in this scenario.
- Analyze the best, average, and worst-case complexities.
- Justify whether sorting would improve system efficiency.

Code of Conduct

I Y Greeshma Reddy (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: Y Greeshma Reddy

RUBRICS FOR CALCULATION

Criteria	Wt.	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Problem Understanding & Context Analysis	25	Thoroughly understands the scenario with accurate interpretation of all requirements	Clearly understands the scenario with minor gaps	Basic understanding; some aspects of the scenario unclear	Poor understanding or misinterpretation of the scenario
Application of Concepts	30	Applies appropriate concepts effectively to solve complex real-world problems	Applies relevant concepts correctly with minor errors	Applies basic concepts but with limited accuracy	Fails to apply appropriate concepts
Analytical & Decision-Making Skills	20	Demonstrates strong analysis and selects optimal solutions with justification	Good analysis with reasonable solutions	Limited analysis; solutions lack justification	Poor or no analysis; incorrect decisions
Solution Design & Approach	15	Well-structured, innovative, and feasible solution approach	Logical and structured solution with minor gaps	Basic solution with limited structure or feasibility	Unclear or incorrect solution approach
Clarity, Justification & Presentation	10	Clear, well-justified explanation with strong reasoning	Clear explanation with some justification	Limited clarity and weak justification	Poorly explained with no justification

Marks Evaluation:

Question No.	Problem Understanding & Context Analysis (25%)	Application of Concepts (30%)	Analytical & Decision-Making Skills (20%)	Solution Design & Approach (15%)	Clarity, Justification & Presentation (10%)	Total Marks (Each – 50)
Q1	4	3	3	2	4	40
Q2	4	4	4	3	2	46
Overall Total (100)						86

Faculty In-charge	Course Coordinator	Head of Department
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

Q1. A library system stores book IDs in an unsorted list due to frequent additions and deletions.

a) Explain how Linear Search is applied in this scenario.

Problem Understanding & Context Analysis

The library stores book IDs in an **unsorted list** because books are frequently added and removed. When a user searches for a book, the system needs a method that works without sorting the data.

Application of Concepts

Linear Search checks each book ID one by one from the beginning of the list until the required book ID is found or the list ends.

Analytical & Decision-Making Skills

Since the list is unsorted, Binary Search cannot be used. Linear Search is the most suitable method because it does not require sorted data.

Solution Design & Approach

1. Start from the first book ID.
2. Compare it with the required book ID.
3. If it matches, stop the search.
4. Otherwise, move to the next book ID.
5. Repeat until the book is found or the list ends.

Clarity, Justification & Presentation

Linear Search is simple, easy to implement, and suitable for frequently updated unsorted data.

b) Analyze the best, average, and worst-case complexities.

Problem Understanding & Context Analysis

The search time depends on where the required book ID is located in the list.

Application of Concepts

Time complexity measures the number of comparisons required to find the book.

Analytical & Decision-Making Skills

- **Best Case: $O(1)$** – Book found at the first position.
- **Average Case: $O(n)$** – Book found around the middle.
- **Worst Case: $O(n)$** – Book found at the last position or not found.

Solution Design & Approach

The algorithm compares one book ID at a time until a match is found or all elements are checked.

Clarity, Justification & Presentation

Linear Search has a linear time complexity because the number of comparisons increases as the number of books increases.

c) Justify whether sorting would improve system efficiency.

Problem Understanding & Context Analysis

The library performs frequent insertions and deletions, making it difficult to maintain sorted data.

Application of Concepts

Sorting allows the use of **Binary Search**, which searches much faster with **$O(\log n)$** complexity.

Analytical & Decision-Making Skills

Sorting improves search speed but increases the time needed for inserting and deleting books.

Solution Design & Approach

- Use **Linear Search** if updates are frequent.
- Use **Sorting + Binary Search** if searches are frequent and updates are rare.

Clarity, Justification & Presentation

Sorting improves search efficiency, but for a library with frequent updates, Linear Search is a more practical and efficient choice.